

Chapter 7: Attitude Representation & Attitude Dynamics – week 2

Dr. Alexander Wittig

- Any questions on previous weeks content?

- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

- **Understand how $\omega(t)$ evolves with time**

- **Rigid Body Mechanics/Dynamics**
 - Center of mass
 - Angular momentum vector
 - Inertia matrix
 - Kinetic energy

- **Dynamics**
 - Torques
 - Euler's equation

Rigid body properties

Center of Mass

- Mass-weighted average of positions
- For discrete (compound) objects:

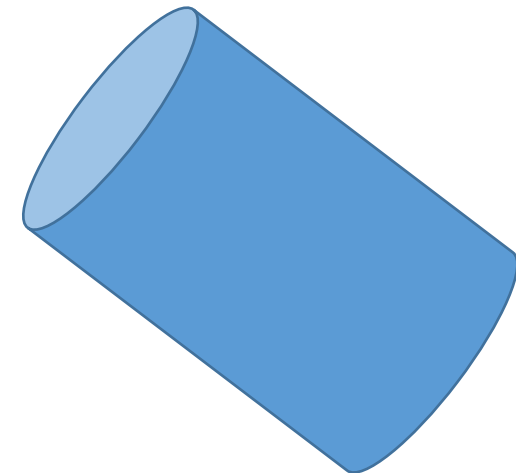
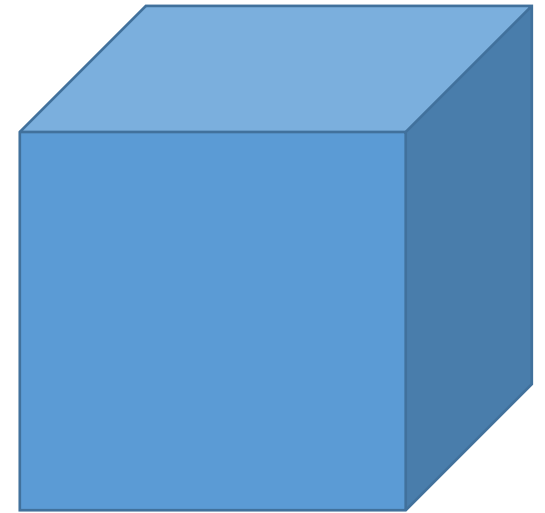
$$\vec{x}_{cm} = \frac{\sum_i m_i \vec{x}_i}{\sum_i m_i}$$

- For continuous mass distributions

$$\vec{x}_{cm} = \frac{\int_V \rho(\vec{x}) \vec{x} dV}{\int_V \rho(\vec{x}) dV}$$

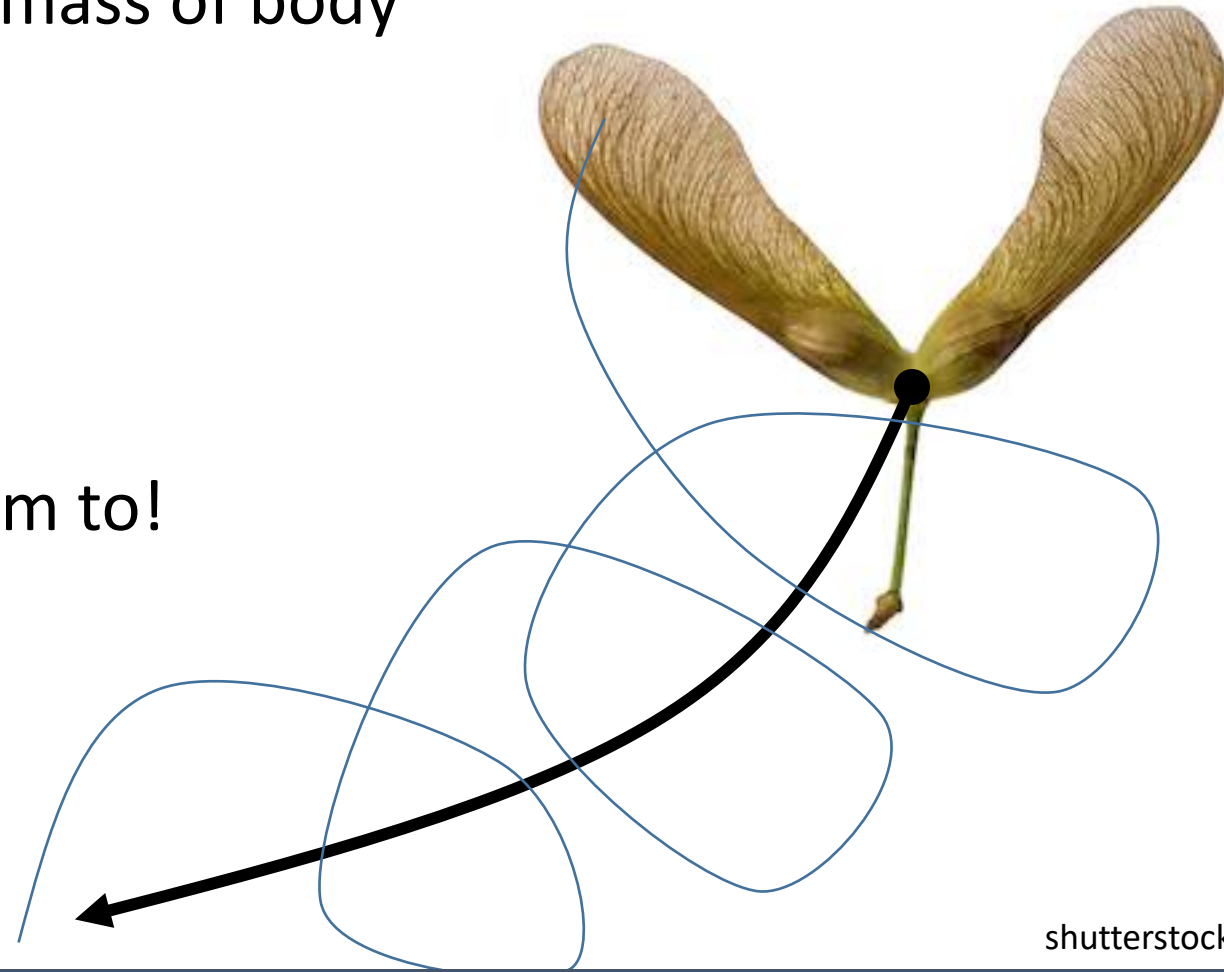
- Often by inspection from symmetry

total mass



Properties

- Only “special” point in rigid bodies
 - Moves linearly like point mass with full mass of body
 - Only affected by **external** forces
 - Not affected by rotational state (except where it affects external forces)
 - Natural pivot point
- ➔ Great point to attach coordinate system to!
- Solve linear motion of center of mass as a point mass
 - Solve rotational motion around center of mass



shutterstock

Angular Momentum Vector

- Conserved quantity in any **inertial frame**
- ALWAYS relative to a reference point (origin) in inertial frame!
- Sum up total angular momentum for rigid body

$$\vec{H} = \sum_i m_i (\vec{r}_i \times \vec{v}_i) \quad \text{or} \quad \vec{H} = \int_V \rho (\vec{r} \times \vec{v}) dV$$

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- ALWAYS relative to a reference point (origin) in inertial frame!
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NOTES:

- **H** body angular momentum with mass
(vs reduced orbital angular momentum **h**)
- Difficult integral, indirectly depends on time (through **r** and **v**)
- **v** refers to the velocities observed in the inertial frame

$$\vec{H} = \sum_i m_i (\vec{r}_i \times \vec{v}_i) \qquad \vec{H} = \int_V \rho (\vec{r} \times \vec{v}) dV$$

can be shown to be identical to:

$$\vec{H} = \mathbf{I} \vec{\omega}$$

where

$$\begin{aligned} \mathbf{I} &= \int_V \rho (r^2 \mathbf{E} - \vec{r} \vec{r}^T) dV \\ &= \int_V \rho \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix} dV \end{aligned}$$

$$\vec{H} = \sum_i m_i (\vec{r}_i \times \vec{v}_i)$$

$$\vec{H} = \int_V \rho (\vec{r} \times \vec{v}) dV$$

can be shown to be identical to:

where

$$\vec{H} = \mathbf{I} \vec{\omega}$$

$$\mathbf{I} = \int_V \rho \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix} dV$$

see Chapter 7 Video Lecture 29

Rigid body properties

Inertia Matrix

Angular momentum and angular velocity are related by simple matrix **I**

$$\mathbf{I} = \int_V \rho (r^2 \mathbf{E} - \vec{r} \vec{r}^T) dV$$

$$= \int_V \rho \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix} dV$$

Matrix of Inertia or Inertia Matrix or Inertia Tensor

- depends on coordinate system and body orientation
- can be relative to any base point and basis
- only valid for rotations around that base point

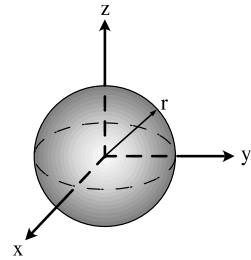
- Inertia matrix is **symmetric**
 - 3 real eigenvalues
 - mutually orthogonal eigenvectors
- Eigenvectors form coordinate basis
 - basis vectors are **principal axes**
 - eigenvalues are **principal moments**
- In principal axes basis, inertia matrix is diagonal!

$$\begin{aligned} \mathbf{I} &= \int_V \rho (r^2 \mathbf{E} - \vec{r} \vec{r}^T) dV \\ &= \int_V \rho \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix} dV \end{aligned}$$

➔ Best coordinate system to use:

- centered at center of mass (for simple linear motion)
- basis along principal body axes (for simple inertia matrix)

Common Inertia Matrices

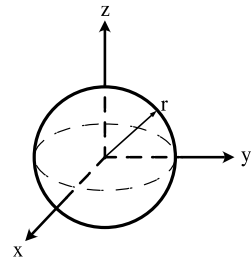


uniform solid sphere
(center of mass)

$$I_x = \frac{2}{5}mr^2$$

$$I_y = \frac{2}{5}mr^2$$

$$I_z = \frac{2}{5}mr^2$$

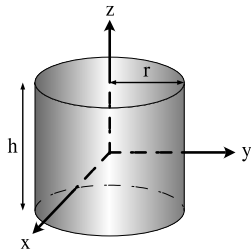


uniform spherical shell
(center of mass)

$$I_x = \frac{2}{3}mr^2$$

$$I_y = \frac{2}{3}mr^2$$

$$I_z = \frac{2}{3}mr^2$$

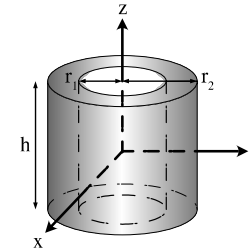


uniform solid cylinder
(center of mass)

$$I_x = \frac{1}{12}m(3r^2 + h^2)$$

$$I_y = \frac{1}{12}m(3r^2 + h^2)$$

$$I_z = \frac{1}{2}mr^2$$

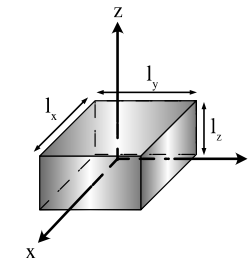


uniform hollow cylinder
(center of mass)

$$I_x = \frac{1}{12}m(3(r_1^2 + r_2^2) + h^2)$$

$$I_y = \frac{1}{12}m(3(r_1^2 + r_2^2) + h^2)$$

$$I_z = \frac{1}{2}m(r_1^2 + r_2^2)$$

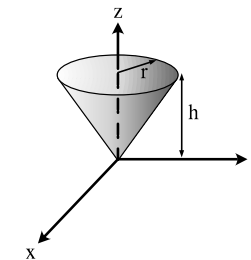


uniform solid cuboid
(center of mass)

$$I_x = \frac{1}{12}m(l_y^2 + l_z^2)$$

$$I_y = \frac{1}{12}m(l_x^2 + l_z^2)$$

$$I_z = \frac{1}{12}m(l_x^2 + l_y^2)$$



uniform solid cone
(apex)

$$I_x = \frac{3}{5}mh^2 + \frac{3}{20}mr^2$$

$$I_y = \frac{3}{5}mh^2 + \frac{3}{20}mr^2$$

$$I_z = \frac{3}{10}mr^2$$

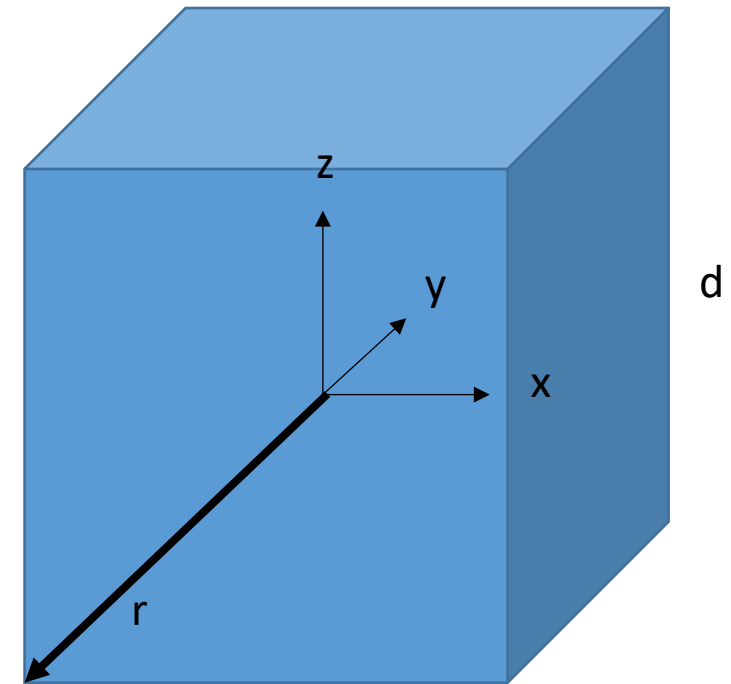
- All relative to center of mass (except for the cone)
- Basis as shown

To shift inertia matrix at center of mass to point $\mathbf{r}$

$$\mathbf{I}_r = \mathbf{I}_{cm} + m(\vec{r}^2 \mathbf{E} - \vec{r}\vec{r}^T)$$

where m is the total mass of the body
(generalized parallel axis theorem)

- Like all mass concentrated at point $\mathbf{r}$
- Useful if rotation is forced around different point



Rotation of Inertia Matrix

To change Inertia Matrix from one coordinate basis (**E**) to another (**e**)

- Rotation matrix **R** such that:

$$\vec{x}_e = \mathbf{R} \cdot \vec{x}_E$$

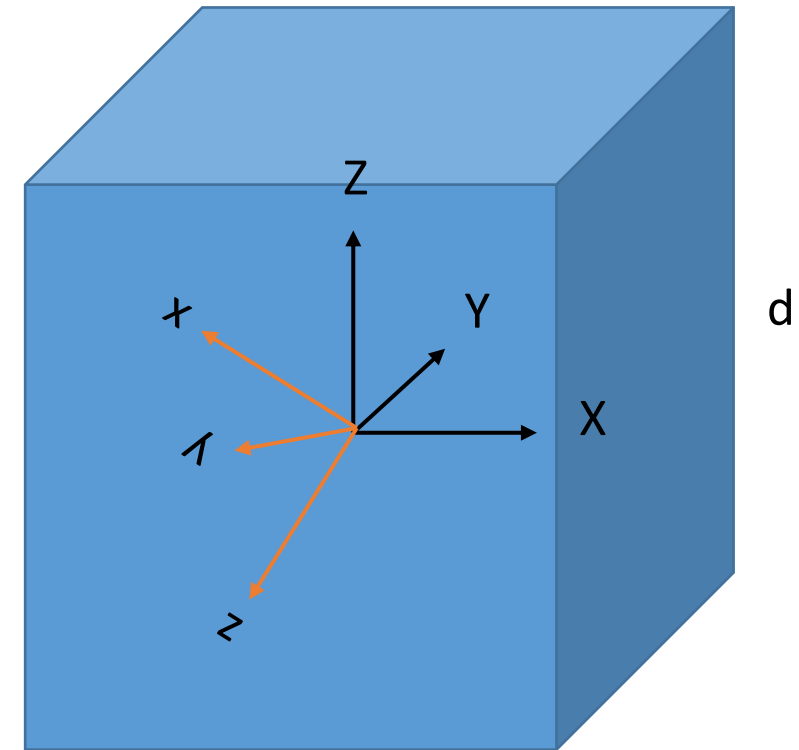
- Then: ← applied right to left

$$\mathbf{I}_e = \mathbf{R} \cdot \mathbf{I}_E \cdot \mathbf{R}^T$$

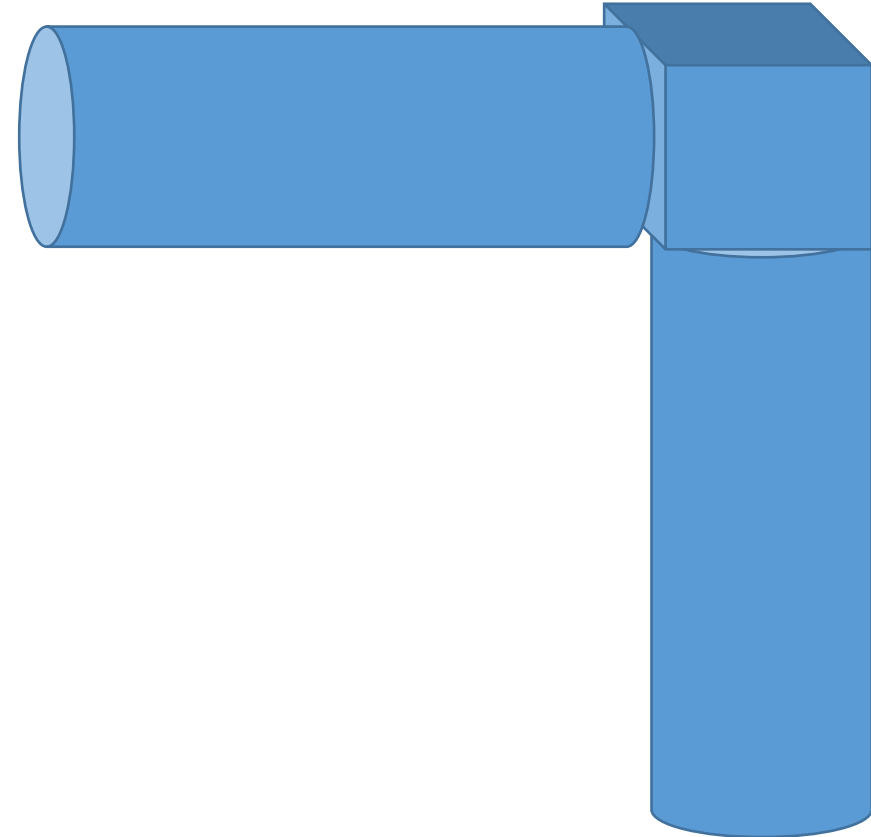
Rotate E
back to e
apply inertia
matrix in E
rotate e
into E

Useful when:

- assembling objects from parts
- the object rotates relative to basis



- Inertia Matrix is additive...
 - ... when relative to **same origin**
 - ... when relative to **same basis**
- Can build composite objects from simple, known shapes



Recipe:

1. Find common reference point (e.g. CoM)
2. Calculate each Inertia Matrix
3. Shift reference point and rotate axes as needed
4. Sum all components

- Inertia Matrix is additive...
 - ... when relative to **same origin**
 - ... when relative to **same basis**
- Can build composite objects from simple, known shapes

Recipe:

1. Find centre of mass point (e.g. CoM)
2. Calculate inertia Matrix
3. Shift reference point and rotate axes as needed
4. Sum all components



Kinetic Energy

- Body motion split into
 - linear motion of center of mass
 - rotational motion around center of mass
- Kinetic energy associated with each
 - linear: $\frac{1}{2}m_{total}\vec{v}_{CM}^2 = \frac{1}{2}\vec{v}_{CM}^T m_{total} \vec{v}_{CM}$
 - rotational: $\frac{1}{2}\vec{\omega}^T \mathbf{I} \vec{\omega}$
- Total kinetic energy: sum of both

Satellite of mass 1500 kg in 300 km circular Earth orbit.

$$\mu_{Earth} = 3.986 \cdot 10^{14} \text{ m}^3/\text{s}^2$$

$$R_{Earth} = 6378 \text{ km}$$

Inertia matrix relative to center of mass and angular velocity:

$$\mathbf{I} = \begin{pmatrix} 2000 & 0 & 0 \\ 0 & 3000 & 0 \\ 0 & 0 & 4000 \end{pmatrix} \text{ kg m}^2$$

$$\vec{\omega} = \begin{pmatrix} 1 \\ -0.9 \\ 1.5 \end{pmatrix} \text{ rad/s}$$

What is its total kinetic energy?

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What is its total kinetic energy?

For circular orbits: $v = \sqrt{\frac{\mu}{r}}$

$$\rightarrow T = 44766 \text{ MJ} + 6715 \text{ J} \approx 44766 \text{ MJ}$$

Clearly dominated by orbital velocity: Attitude maneuvers are cheap!

Euler's Equation & Torques

- In **inertial frame** equations of motion (dynamics) are particularly simple

$$\frac{d}{dt}\vec{p} = \frac{d}{dt}m\vec{v} = \vec{F}$$

linear motion (Newton's second law)

$$\frac{d}{dt}\vec{H} = \frac{d}{dt}\mathbf{I}\vec{\omega} = \vec{T}$$

angular motion

- In **inertial frame** equations of motion (dynamics) are particularly simple

$$\frac{d}{dt}\vec{p} = \frac{d}{dt}m\vec{v} = \boxed{\vec{F}} \quad \text{external forces} \quad \text{linear motion (Newton's second law)}$$
$$\frac{d}{dt}\vec{H} = \frac{d}{dt}\mathbf{I}\vec{\omega} = \boxed{\vec{T}} \quad \text{external torques} \quad \text{angular motion}$$

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$$\frac{d}{dt}\vec{H} = \frac{d}{dt}\mathbf{I}\vec{\omega} = \boxed{\vec{T}} \quad \text{external torques} \quad \text{angular motion}$$

An inertial frame

- is moving without accelerating
- is not rotating

- In **inertial frame** equations of motion (dynamics) are particularly simple

$$\frac{d}{dt}\vec{p} = \frac{d}{dt}m\vec{v} = \boxed{\vec{F}} \text{ external forces}$$

linear motion (Newton's second law)

$$\frac{d}{dt}\vec{H} = \frac{d}{dt}\mathbf{I}\vec{\omega} = \boxed{\vec{T}} \text{ external torques}$$

angular motion

An inertial frame

- is moving without accelerating
- is not rotating

but: with respect to what?

- In theory: frame is measured relative to distant stars (“fixed stars”)
- In practice: anything that is “sufficiently” inertial will do:
 - Earth is rotating slowly compared to e.g. spinning top
 - Earth motion around sun is slow compared to low Earth satellite

Example: Foucault’s pendulum



Wikipedia

- Torques from forces: $\vec{T} = \vec{r} \times \vec{F}$
 - measured from center of rotation
- Units: Nm (Newton-meter)
- Torques are additive
 - must be around the same point!
- Torques are “twisting forces”



torque wrench (Capri)

Component	Brand	Newton meters	Inch-pounds
Bearing cups	Shimano® Hollowtech II	34.5~49.1 Nm	305~435 in-lbs
Cartridge bottom brackets	Campagnolo® (3-piece)	70 Nm	612 in-lbs
	Campagnolo® Ultra-Torque cups	35 Nm	310 in lbs
	FSA®	39.2~49 Nm	347~434in lbs
	Race Face®	47.5 Nm	420 in lbs
	Shimano®	49~69 Nm	435~610 in lbs
	Truvativ®	33.9~40.1 Nm	300~360 in lbs
	White Industries®	27 Nm	240 in lbs

Size	Class 4.8				Class 8.8 or 9.8				Class 10.9				Class 12.9			
	Lubricated ^a		Dry ^a		Lubricated ^a		Dry ^a		Lubricated ^a		Dry ^a		Lubricated ^a		Dry ^a	
	N-m	lb-ft	N-m	lb-ft	N-m	lb-ft	N-m	lb-ft	N-m	lb-ft	N-m	lb-ft	N-m	lb-ft	N-m	lb-ft
M6	4.8	3.5	6	4.5	9	6.5	11	8.5	13	9.5	17	12	15	11.5	19	14.5
M8	12	8.5	15	11	22	16	28	20	32	24	40	30	37	28	47	35
M10	23	17	29	21	43	32	55	40	63	47	80	60	75	55	95	70
M12	40	29	50	37	75	55	95	70	110	80	140	105	130	95	165	120
M14	63	47	80	60	120	88	150	110	175	130	225	165	205	150	260	190
M16	100	73	125	92	190	140	240	175	275	200	350	225	320	240	400	300
M18	135	100	175	125	260	195	330	250	375	275	475	350	440	325	560	410
M20	190	140	240	180	375	275	475	350	530	400	675	500	625	460	800	580
M22	260	190	330	250	510	375	650	475	725	540	925	675	850	625	1075	800
M24	330	250	425	310	650	475	825	600	925	675	1150	850	1075	800	1350	1000
M27	490	360	625	450	950	700	1200	875	1350	1000	1700	1250	1600	1150	2000	1500
M30	675	490	850	625	1300	950	1650	1200	1850	1350	2300	1700	2150	1600	2700	2000
M33	900	675	1150	850	1750	1300	220	1650	2500	1850	3150	2350	2900	2150	3700	2750
M36	1150	850	1450	1075	2250	1650	2850	2100	3200	2350	4050	3000	3750	2750	4750	3500

- In inertial (rest) frame:

 $\vec{X}, \vec{Y}, \vec{Z}$

$$\frac{d}{dt}\vec{H} = \vec{T}$$

- In frame rotating with angular velocity $\vec{\omega}$ (e.g. body fixed frame)

 $\vec{x}, \vec{y}, \vec{z}$

$$\frac{d}{dt}\vec{H} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

- In inertial (rest) frame:

$$\vec{X}, \vec{Y}, \vec{Z}$$

$$\frac{d}{dt}\vec{H} = \vec{T}$$

- In frame rotating with angular velocity $\vec{\omega}$ (e.g. body fixed frame)

$$\vec{x}, \vec{y}, \vec{z}$$

$$\frac{d}{dt}\vec{H} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

see Chapter 7 Video Lecture 31

$$\frac{d}{dt} \vec{H} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

- $\left. \frac{d}{dt} \vec{H}, \frac{d}{dt} \vec{H} \right|_{inertial}$ derivative of vector as observed in inertial frame,
(can be expressed in the rotating frame coordinates)
- $\left. \dot{\vec{H}}, \frac{d}{dt} \vec{H} \right|_{rot}$ derivative of vector in rotating frame,
i.e. the derivative of its components expressed in
rotating frame coordinates

These are, in general, different

Valid in general: $\frac{d}{dt}\vec{a} = \dot{\vec{a}} + \vec{\omega} \times \vec{a}$

Example: angular acceleration

$$\vec{\alpha} = \frac{d}{dt}\vec{\omega} \quad \text{in inertial frame (by definition)}$$

$$\vec{\alpha} = \dot{\vec{\omega}} + \vec{\omega} \times \vec{\omega} = \dot{\vec{\omega}} \quad \text{in rotating frame (by above relation)}$$

- Special case! $\frac{d}{dt}\vec{\omega} = \dot{\vec{\omega}}$

Euler's equation in body-fixed frame

$$\frac{d}{dt}\vec{H} = \vec{T}$$

$$\frac{d}{dt}\vec{H} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

$$\vec{\omega} = \mathbf{I}^{-1} \vec{H}$$

Putting it all together:

Rate of change seen
in body fixed frame

Angular velocity of
body-fixed frame
relative to inertial frame

$$\begin{aligned} \dot{\vec{H}} &= \vec{T} - \vec{\omega} \times \vec{H} \\ &= \vec{T} - \left(\mathbf{I}^{-1} \vec{H} \right) \times \vec{H} \end{aligned}$$

Total external
torques

Full equations of motion in body fixed frame

$$\dot{\vec{H}} = \vec{T} - \vec{\omega} \times \vec{H}$$

angular momentum

$$\dot{\vec{q}} = \frac{1}{2} \begin{pmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{pmatrix} \vec{q}$$

Quaternion
(orientation)

*from last week

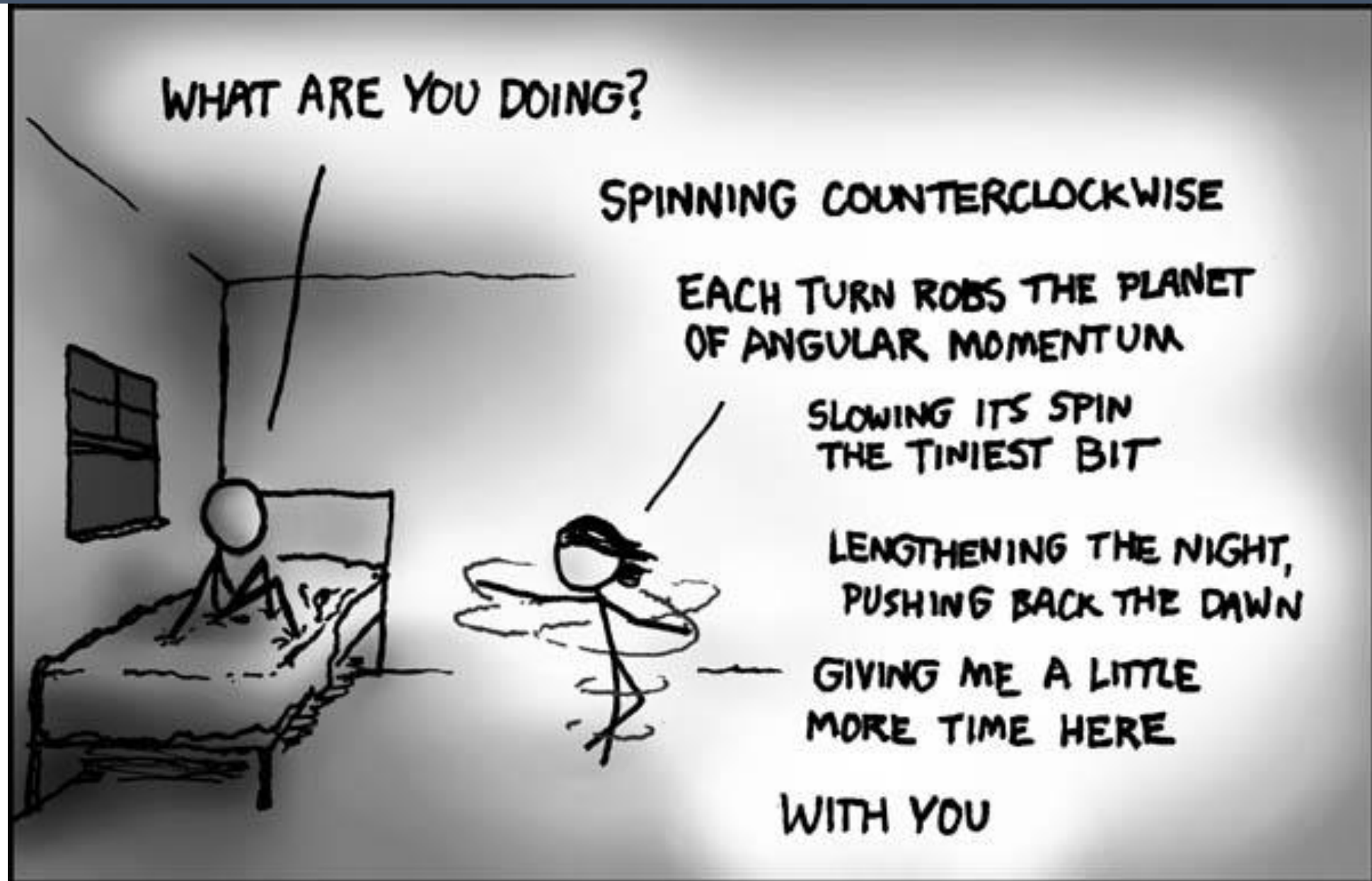
where $\vec{\omega} = \mathbf{I}^{-1} \vec{H}$ and $\mathbf{I}$ is constant.

- Body properties
 - Center of Mass
 - Inertia Matrix
 - Principal axes
- Mechanics
 - Angular Momentum Vector
 - Kinetic Energy
- Dynamics
 - Euler's Equation
 - Torques

Further reading:

Curtis: 9.3 – 9.8

Chobotov: 1.3



Chapter 7: Attitude Representation & Attitude Dynamics – week 2

Dr. Alexander Wittig